

Introductory Chapter: Water Quality – Emerging Concern for Sustainable Future

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1. Water quality in human history and its contemporary state

Water quality has not been in focus for a very long period of human history on our planet. Most probably, there have been many occasions when humans have learned not to drink “dirty” (toxic, polluted, infected) waters due to the impact on health or even death, which was firmly embedded in the collective consciousness. As the Bible states [1], even God has used toxic waters to punish the Egyptians by turning the waters of the River Nile red.

After so many millennia of development and progress, human society has unfortunately reached a very grim state regarding global water quality. It is dimly expressed [2] as: *“In many countries, both developing and developed, current pathways for water use are often not sustainable. . . The world faces a worsening series of local and regional water quantity and quality problems. . . Water resource constraints and water degradation are weakening one of the resource bases on which human society is built.”* [2] The statement is so profoundly corroborated by the newest facts [3] based on scientific research, which can be summarized as:

- (1) Only about 44% of global household wastewater is safely treated before being released, leaving a majority that is discharged without adequate treatment [4].
- (2) Less than half of industrial wastewater is safely treated globally, contributing to chemical, nutrient, and pathogen contamination of rivers and lakes [4].
- (3) By 2030, the health and livelihoods of up to 4.8 billion people could be at risk if water quality monitoring and treatment are not improved [4].
- (4) Roughly half of the world’s population experiences severe water scarcity for at least part of the year, and water pollution aggravates these shortages by reducing usable freshwater [5].
- (5) In lower-income countries, poor water quality is mainly due to low levels of wastewater treatment, whereas in higher-income countries, agricultural runoff is a leading pollution source [5].

- (6) Emerging water contaminants include pharmaceuticals, hormones, industrial chemicals, detergents, and perfluoroalkyl and polyfluoroalkyl substances (PFAS) – substances that are persistent in the environment and difficult to remove [5].
- (7) Wastewater often carries mixtures of pollutants – including microplastics, pathogens, heavy metals, and chemicals – that can disrupt aquatic food webs and affect wildlife [6].
- (8) Persistent, toxic “forever chemicals” (PFAS) are increasingly found in US tap water and are linked to cancer, immune system harm, and other serious health risks [7].
- (9) Groundwater contamination from naturally occurring geologic elements – like arsenic, fluoride, selenium, and uranium – affects drinking water safety around the world. In some regions, a significant share of groundwater sources exceeds recommended guidelines [8].
- (10) Global water quality is declining while water demand increases, requiring better wastewater management, pollution reduction, and monitoring to protect human and environmental health [5].

Many water quality indicators vary with flow conditions and over time. Without consistent monitoring, short-duration pollution events may be missed, delaying response and increasing impacts on ecosystems and drinking water supplies.

2. The marine environment

Moreover, the situation with the marine environment seems to be at least equally alarming, if not more so. Academics have already sounded the alarm about the destruction of global ecosystems [9]. Humanity must immediately and significantly alter our harmful trajectory in order to avoid irrevocably damaging our oceans in multiple ways that will further affect ocean health for both us and future generations. The ocean is the world’s largest realm, housing an astonishing array of biodiversity that provides critical ecological functions that ultimately support life on Earth [10].

Humanity relies on the ocean for livelihood and wellbeing, including a variety of benefits ranging from food production, tourism, pharmaceuticals, energy production, shipping, recreation, esthetics, culture, and scientific research. Although these services are not always quantifiable, humans heavily depend on an immeasurable number of services provided by a healthy ocean, including shoreline protection, nutrient and carbon cycling, climate and weather regulation, oxygen production, primary productivity, waste detoxification, and biodiversity maintenance [11]. Placing a global value on marine ecosystem services is difficult or even impossible, likely leading us to make the dangerous mistake of undervaluing the ocean despite the large body of evidence that a healthy ocean is critical to the success of all life on the planet.

Approximately 98% of the global ocean is already affected by multiple stressors, raising the concerning possibility of large-scale ecological collapse in a number of systems. While coastal habitats often experience the most significant damage, even remote offshore or deep-water habitats are not immune to increasing threats from human activities. Damage to marine habitats has accelerated rapidly around the world, with 76% of marine and estuarine species affected, including notable losses of 50% of coral-reef cover [12] and 40% of kelp forests. Simultaneously, extensive biodiversity loss is occurring; populations of marine vertebrates decreased by an average of 22% from 1970 to 2008, with marine fish experiencing a higher decrease of 38%. More than 25 marine species have already been driven extinct, with extinction rates expected to increase sharply in the near future. In addition to biodiversity and habitat losses, human disruptions render marine ecosystems less resilient, considerably increasing their vulnerability to further disturbances [13].

3. Selected papers regarding water quality in this book

Many interesting works of high quality have been received during the preparation of this book. Especially notable are many papers coming from scientists who work in developing countries, who have meticulously documented harmful impacts on water quality and the consequent deleterious effects on their societies. While these attempts are quite commendable for creating awareness of the dangers linked to the deterioration of water quality for human health and the environment, this book seeks to present some novel aspects in monitoring, new methods, a comprehensive approach, or new technologies for the treatment and prevention of water pollution. And indeed, this book's title has attracted some quite interesting research topics like contaminants of emerging concern, hydrodynamic cavitation technology for the degradation of organic pollutants, or a dual-wavelength continuous-flow nephelometer, just to name a few.

4. Conclusions

It is evident that the global scientific community repeatedly and constantly points out the dangers of water quality deterioration and the urgent need for rapid, decisive actions to reverse the already deleterious effects. The current issue on water quality attempts to, once again, initiate the responsibility of the modern human society toward sustainable practice regarding water quality, but also overall and global care for Planet Earth. Through a modern approach and new insights, we hope to achieve strong momentum among decision makers to initiate much more intensive rehabilitation, prevention, and restoration activities in order to preserve water environments, human health, and biodiversity for future generations.

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